

Rohan Devraj

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EDUCATION

Georgia Institute of Technology | Atlanta, GA

Aug. 2023 - May 2027

B.S. in Industrial & Systems Engineering, Minor in Computer Science

GPA: 3.84

- *Specializations:* Operations Research, Artificial Intelligence & Machine Learning
- *Relevant Coursework:* Applied Probability, Optimization, Statistics, Data Input & Manipulation, Discrete Math, Data Structures & Algorithms, Stochastic Systems, Database Systems
- *Clubs/Activities:* 180 Degrees Consulting, Aarohi Indian Classical Arts, The Tower Undergraduate Research Journal, GT Venture Capital Club, Grand Challenges LLC, Intramural Basketball

EXPERIENCE

Optum Insights - Analytics (UnitedHealth Group)

Jun. 2025 – Aug. 2025

Data Engineering Intern

Minneapolis, MN

- Developing a Spark-compatible Python library deployed in Azure Databricks to compute AHRQ Quality & Patient Safety measures from FHIR-formatted clinical data, replacing legacy SAS systems and saving \$36K annually
- Reducing processing time by 90% (10 hrs to < 1 hr) by optimizing with Pandas UDFs and Databricks
- Built validation pipelines to benchmark Spark outputs against legacy SAS logic, ensuring compliance with AHRQ healthcare quality specifications and accelerating clinical data modernization through fast, in-place processing

Space Systems Optimization Group at Georgia Tech

Aug. 2024 – present

Undergraduate Researcher

Atlanta, GA

- Developed and optimized a Mixed Integer Linear Program (MILP) in Python with Gurobi for vehicle routing, modeling 5,340 mapped locations with elevation-based path costs (advised by Dr. Koki Ho)
- Solved the model with 52,000+ variables and 62,000+ constraints, achieving optimal route plan in under 8 seconds
- Led research on adaptive path planning by integrating local and global routing to improve dynamic decisions under satellite and terrain data uncertainty

Carnegie Mellon Department of Statistics & Data Science

Jun. 2024 – Aug. 2024

Summer Undergraduate Research Fellow | stat.cmu.edu/cmsac/sure/2024/showcase/

Pittsburgh, PA

- Nationally selected (1 of 18) for a competitive research program in data science applied to healthcare analytics
- Developed regression and machine learning models (linear, lasso, ridge, decision trees) to analyze the impact of adult health behaviors on child mortality and low birthweight at the county level
- Evaluated and compared models using 10-fold cross-validation; selected linear regression for its 5% lower root mean squared error (RMSE), enhancing low birthweight prediction accuracy
- Conducted 10+ analyses on maternal healthcare disparities and applied k-means clustering to identify distinct demographic and behavioral risk groups

Autonomous & Connected Transportation Laboratory at Georgia Tech

Jan. 2024 – May. 2024

Undergraduate Research Assistant

Atlanta, GA

- Formulated a multi-objective traffic assignment model in Python using Pyomo and Gurobi to optimize routing across zone-based transportation networks, balancing travel time, congestion, and safety across diverse zones
- Minimized pairwise accessibility disparities across zones to promote fair distribution of travel opportunities

LEADERSHIP

Georgia Tech Alumni Association Angel Network

Aug. 2024 – present

Associate | gtangelnetwork.com

Atlanta, GA

- Lead Georgia Tech's first angel investment network, sourcing 20+ startups from a 100+ deal pipeline and supporting over \$6.5M in connected capital
- Direct meetings with founders and collaborate with investors to craft detailed investment memos, analyzing product, market strategy, financial performance, and growth potential

SKILLS

Programming/Tools: Python, SQL, Java, R, Excel, PySpark, Azure Databricks, SAS, Git, Copilot

Libraries & Modeling: Pandas, NumPy, ggplot2, dplyr, Gurobi, Pyomo, GurobiPy